

TYPED CONTROL, INFORMATION, AND SADDLE ENVELOPES FOR KINETIC COTANGENT PHASES

QIAN QI

ABSTRACT. The exact information state is a positive posterior law obtained by integrated coarea/Rokhlin disintegration; oriented currents retain only geometry. A compact evidence-direction cone closes the zero-evidence boundary, and the full-belief controlled kernel satisfies a Feller DPP. A growing moment coordinate has a quantitative Wasserstein error, while coefficient-first inserted local limits yield controlled LAN and Bernstein–von Mises only on uniformly informative regular strategies.

Platform identifier: HSBG-CONTROL-v5. **Controlling revision:** ROUND13-REFEREE-POSITIVE-CLOSURE. **Registered proof source:** C1_DISINTEGRATED_BELIEF_GROWING_MOMENTS.tex.

CONTENTS

1. Round-thirteen positive closure: disintegrated belief kernels and growing-moment reduction	1
1.1. Positive stratified disintegration	2
1.2. The evidence cone	2
1.3. Exact full-belief dynamic programming	3
1.4. A quantitative growing-moment reduction	3
1.5. Inserted coefficients and regular statistical limits	4
Scope and verification status	5

1. ROUND-THIRTEEN POSITIVE CLOSURE: DISINTEGRATED BELIEF KERNELS AND GROWING-MOMENT REDUCTION

The exact information state is a probability law, not an oriented current. Currents retain collision and observation geometry, while positive Rokhlin slices supply evidence and posterior probabilities. No fixed finite list of moments is declared sufficient: the reduced coordinate order grows with the Boltzmann–Grad scale and has an explicit approximation error.

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 49L25, 93E20; Secondary 60G35, 82C40.

Key words and phrases. risk-sensitive control, kinetic equation, filtering, saddle envelope, deterministic information.

1.1. Positive stratified disintegration. Let X be the graph-completed microscopic/history state and let $O_u : X \rightarrow Y_u$ be one of a finite family of C^2 observation maps. On every Whitney stratum S , $O_u|_S$ has constant rank. For a positive finite measure ν whose stratum densities belong to the declared weighted Sobolev class, coarea/Rokhlin disintegration gives

$$\nu(dx) = \int_{Y_u} \nu_{u,y}(dx) Z_\nu^u(y) dy, \quad \nu_{u,y}(X) = 1$$

after omitting the zero-density fibers. The oriented slice current and the positive coefficient measure $Z_\nu^u(y)\nu_{u,y}$ are stored separately.

Define the observation kernel as the probability measure on $Y_u \times \mathcal{P}_W(X)$

$$\mathbf{O}_u(\nu)(dy, d\pi) = Z_\nu^u(y) dy \delta_{\nu_{u,y}}(d\pi).$$

Theorem 1.1 (Feller disintegrated observation kernel). *On a compact regular stratum chart with uniform submersion, density, and weighted-tail bounds, the map $\nu \mapsto \mathbf{O}_u(\nu)$ is weakly continuous. Hence, for every bounded continuous posterior payoff Φ ,*

$$\nu \mapsto \int_{Y_u} \Phi(\nu_{u,y}) Z_\nu^u(y) dy$$

after including the observed reward is continuous. Critical-rank strata are included as separate coarea components.

Proof. Use a finite family of constant-rank charts and a partition of unity. In each chart the coarea formula writes the joint observation/posterior integral as an ordinary integral over graph coordinates with a uniformly bounded Jacobian. Weak convergence plus the uniform Sobolev/tail bounds gives convergence of these integrals for a determining algebra on $Y_u \times \mathcal{P}_W(X)$. Tightness upgrades it to weak convergence of the joint kernel. Summing the strata proves the theorem. No pointwise continuity of an arbitrarily chosen version of $y \mapsto \nu_{u,y}$ is asserted. $\square$

1.2. The evidence cone. Let $\mathcal{K}_M \subset \mathcal{P}_W(X)$ be a compact weighted belief set. The compactified unnormalized state is

$$\widehat{\mathfrak{B}}_M = \{(r, \pi) : 0 \leq r \leq 1, \pi \in \mathcal{K}_M\} / \sim,$$

where (r, π) represents the positive measure $r\pi$, and at $r = 0$ the projective direction π is retained. On each regular blow-up chart the direction is the normalized leading coarea coefficient.

Proposition 1.2 (Continuous zero-evidence lift). *Prediction followed by the joint kernel of Theorem 1.1 extends continuously to $\widehat{\mathfrak{B}}_M$. A zero-evidence observation has zero probability; its retained direction is used only to make subsequent unnormalized transitions continuous and does not alter any expected value.*

Proof. Write the unnormalized sliced measure as $r^k G(u, u, y) + o(r^k)$ in a fixed constant-rank blow-up chart and normalize the first nonzero positive coefficient. The regularity assumptions make this coefficient unique and

continuous. If all coefficients vanish, use the predicted direction, which is likewise continuous. Since the observation kernel weights the state by its evidence r^k , the chosen zero-mass direction does not change the integrated transition. The joint-kernel continuity then proves the extension. $\square$

1.3. Exact full-belief dynamic programming. Let $\mathfrak{B}$ be the space of complete posterior laws. A control first chooses a canonical actual-contact likelihood as in C1's microscopic game, then the physical law is propagated, and finally the new observation is disintegrated by $\mathbf{O}_u$.

Theorem 1.3 (Exact belief-state DPP). *The prediction–observation kernels and the risk-sensitive reward define a Feller controlled Markov kernel on $\mathfrak{B}_M$. Nonanticipative relaxed strategies satisfy the exact dynamic programming principle. The one-time microcanonical preparation variable is chosen once and retained as a static coordinate; it is not reoptimized after every observation.*

Proof. Ionescu–Tulcea constructs the controlled path law from the prediction kernel and the joint disintegration kernel. Theorem 1.1 and Proposition 1.2 give Feller continuity and compact containment. Conditioning at an observation time and concatenating relaxed strategies gives the DPP. The initial shell multiplier belongs to the initial law and therefore remains fixed under conditional updates. $\square$

1.4. A quantitative growing-moment reduction. For a belief π , let $M_K(\pi)$ be all mixed moments of a fixed separating countable observable family of total degree at most K . Suppose $\int e^{\alpha W} d\pi \leq M$ on the reachable class. Let $K_\varepsilon \rightarrow \infty$ and $R_\varepsilon \rightarrow \infty$ satisfy

$$\frac{R_\varepsilon}{K_\varepsilon} \rightarrow 0, \quad e^{-\alpha R_\varepsilon} = o(1).$$

For example, take $R_\varepsilon = \log \mu_\varepsilon$ and $K_\varepsilon = (\log \mu_\varepsilon)^2$.

Lemma 1.4 (Quantitative moment approximation). *If two beliefs in the reachable exponential-moment class have moment vectors M_{K_ε} differing by $o(K_\varepsilon^{-d})$ in the normalized coordinates, then*

$$W_{1,W}(\pi, \tilde{\pi}) \leq C \frac{R_\varepsilon}{K_\varepsilon} + C e^{-\alpha R_\varepsilon} + o(1).$$

Thus a growing finite coordinate, not a fixed finite cumulant list, is asymptotically determining.

Proof. Truncate to the compact set $\{W \leq R_\varepsilon\}$. Multivariate Bernstein polynomials approximate every unit weighted-Lipschitz test there with error $CR_\varepsilon/K_\varepsilon$. Its polynomial integral is determined by the moments through degree K_ε , and the assumed moment mismatch gives the $o(1)$ term. Exponential moments bound both tails by $Ce^{-\alpha R_\varepsilon}$. Kantorovich duality yields the result. Since moments and cumulants are triangular polynomial transforms at every finite order, either coordinate system may be used. $\square$

Theorem 1.5 (Reduced DPP with a controlled error). *Assume the B2 source chart gives the uniform exponential-moment bound on every belief reachable by the declared controls, and choose K_ε so that the one-step error in Lemma 1.4 is $o(J_\varepsilon^{-1})$, where J_ε is the number of decision stages. Then the full-belief value and the value on $(r, M_{K_\varepsilon}(\pi))$ differ by $o(1)$ uniformly over admissible strategies.*

Proof. Couple one prediction–observation step through the common microscopic law and apply Lemma 1.4 conditionally. Bounded Lipschitz rewards and the Feller kernel make the Bellman operator Lipschitz in weighted Wasserstein distance. Induction over the stages sums at most J_ε one-step errors. The chosen growth of K_ε makes the total $o(1)$. No claim is made that a fixed finite moment vector determines the law. $\square$

1.5. Inserted coefficients and regular statistical limits. Augment the B1 continuous constraint vector by a finite regular observation statistic. Exact number extraction is performed first.

Theorem 1.6 (Inserted observation coefficient). *On a compact regular observation/source chart, the numerator with one bounded posterior cylinder insertion and the denominator satisfy the same exact-number mixed local theorem. Their ratio converges to the tilted conditional expectation, uniformly on regular observation fibers. The large-frequency bound is the integrable canonical coefficient theorem of B1, not a full-grand-canonical atom-free assertion.*

Proof. The insertion changes only the analytic amplitude surrounding the canonical saddle. B1’s coefficient-first Fourier theorem supplies the common Gaussian phase and an integrable relative remainder. The leading Riesz/amplitude coefficient in the numerator is the conditional tilted expectation; division by the positive denominator proves the ratio. $\square$

Let $\vartheta \in \mathbb{R}^d$ be a finite-dimensional identifiable phase parameter. Restrict strategies to the uniformly informative class

$$\frac{1}{J} \sum_{j < J} \mathcal{I}_j(\vartheta_0 \mid \mathcal{F}_{j-1}) \geq cI_d$$

after stopping on a compact regular chart.

Theorem 1.7 (Controlled LAN and Bernstein–von Mises). *For a prior with a positive continuous density near ϑ_0 , the exact finite-centered log likelihood is uniformly LAN over the informative strategy class. Away from phase/coarea boundaries, the posterior of $\sqrt{\mu_\varepsilon}(\vartheta - \vartheta_0)$ converges in total variation to the Gaussian with the random efficient mean and inverse predictable information. No claim is made for uninformative strategies or coexistence boundaries.*

Proof. The inserted coefficient theorem gives a uniform third-order expansion of the conditional log likelihood and exact normalization. The predictable score is a martingale; B3 supplies its joint Gaussian limit, and the uniform

information lower bound gives quadratic coercivity. Exponential tests outside a shrinking neighborhood give posterior tightness. Taylor expansion of the prior and Le Cam's completion-of-squares argument yield total-variation convergence on the regular chart. $\square$

Theorem 1.8 (Round-thirteen C1 closure). *The hard-sphere information problem has an exact Feller full-belief DPP, a positive stratified disintegration with a continuous zero-evidence cone, a quantitative growing-moment reduction, a number-first inserted coefficient, and a controlled LAN/Bernstein-von Mises theorem on the uniformly informative regular strategy class.*

Proof. Combine Theorems 1.1, 1.3, 1.5, 1.6, and 1.7. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn